



Environmentally friendly, robust, and highly efficient

GEAFOL cast-resin transformers for wind turbine
generators

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GEAFOL cast-resin transformers

Immune to vibrations, wind and weather, resistant to salt water, environmentally compatible and vibration-proof: GEAFOL cast-resin transformers for wind turbine generators and offshore wind farms

Cast-resin transformers are mostly used where electricity has to be transformed from the medium-voltage level down to low-voltage, or to another grid voltage. Transformers destined for this function are generally installed close to electrical loads and therefore almost invariably in immediate proximity to people. They must consequently be particularly safe and highly reliable in operation.

Such transformers are found in high-rise buildings, hospitals, data centers, subway stations or on ships and oil production platforms, as well as in paper mills, rolling mills, airports or other locations requiring substantial amounts of power. They are often also installed very close to passers-by and residential areas, separated from the environment only by the stone wall of a building or a metal enclosure. Safety is therefore of utmost priority for everyone concerned.

Wherever distribution transformers have to be installed in areas frequented by people, cast-resin transformers are the preferred option because this particular design ensures users a high level of safety in operation while also offering excellent electrical, mechanical and thermal properties. Transformers with these features require little in the way of maintenance and ensure a high level of safety in terms of behavior in fire. The GEAFOL® cast-resin transformers – a development from the Siemens manufacturing plant in Kirchheim – are reliable in operation, environmentally compatible, flame-retardant, self-extinguishing and produce no toxic gases.

These features are the reason why significantly more than 150,000 of these transformers are in service around the world today, ranging from 50 kVA to approx. 50 MVA and with voltages up to 52 kV, and also explains why licenses were given worldwide to produce GEAFOL transformers.

Predestined for operation in wind turbine generators

All the above attributes also explain why cast-resin transformers from Siemens are so well suited for use in wind turbine generators and offshore wind farms. Wind turbine generators subject transformers to much greater mechanical, electrical and thermal stresses than transformers installed in standardized secondary unit substations. Accessibility is also frequently restricted, which means very high

demands on design, product quality and safety. Cast-resin transformers from Siemens Energy for use in wind turbines are therefore supplied in a mechanically reinforced design, which – when installed in the nacelle – can withstand the strong vibrations without difficulty. These transformers are also painted with a special coating to protect them from the aggressive, salt-laden marine atmosphere. The insulation used on cast-resin transformers is an environmentally sustainable mixture of epoxy resin and quartz powder, which does not produce any toxic gases even under exposure to arcing. Cast-resin transformers are classified according to verified environmental, climate and fire classes. GEAFOL transformers satisfy environmental class E3, climate class C3 and fire class F1 and so conform to the highest requirements¹⁾.

These results are also validated by numerous tests performed by Siemens in collaboration with noted manufacturers of wind turbine generators. These tests are conducted as part of an extensive continuously running program specially designed to test the cast-resin transformers for vibration resistance, lightning impulse withstand capability and short-circuit strength. The results also once again confirm that GEAFOL transformers fulfil the high, by the customers requested, environmental standards and climate classes defined in standard IEC 60076-11. Testing is carried out in well-known European institutes such as KEMA in the Netherlands or IABG in Munich, Germany.



Where high mechanical stresses are encountered, GEAFOL cast-resin transformers of reinforced design are used. This example shows a transformer being used in the nacelle of a wind turbine, where it must reliably withstand constant vibrations.

1) Additional classes upon request

The successful tests for mechanical resistance to operationally induced vibrations and transportation stresses exposed the transformer to accelerations of more than twice gravitational in all three coordinates.

Cast-resin transformers often have a smaller footprint than equivalent liquid-filled or gas-insulated transformers and they are very flexible with respect to the position of the high-voltage/low-voltage terminals. No extra safety precautions, e.g. oil trough, are necessary. Designs with reduced no-load and short-circuit losses, e. g. considering the requirements of the Eco-Design Directive in the EU, increase transformer efficiency and so reduce operating costs. Moreover, epoxy resin-insulated transformers are practically maintenance-free – crucial criteria for offshore wind parks.

Cast-resin transformers in wind turbine generators are used to step up the voltage supplied by the nacelle-mounted generator to the correct level for feeding into the medium-voltage network. Transformers for wind turbine generators are usually designed for operating voltages on the medium-voltage level of up to 40,5 kV respectively up to 72,5 kV for higher ratings, mainly used for offshore installations.



Wind turbine generators make tremendous demands on transformers in terms of electrical and mechanical stressing and consequently also in safety and quality terms. Cast-resin transformers from Siemens Energy for use in wind turbines are therefore supplied mechanically reinforced to ensure vibration-resistant installation in the nacelle.

Further developments afoot

Work on developing cast-resin transformers at Siemens Energy never ceases. Principal Key Expert Bernd Hofmann at the Kirchheim/ Teck factory quotes data sheets which show that the weight of transformers has been reduced by more than one third over the last decades (introduction 1966) alone without any reduction in power rating or any adverse effects on operating safety and reliability. Especially the latest innovation of the GEAFOL cast-resin transformer - the GEAFOL Neo with additional special insulation features - represents an optimal combination of performance, small dimensions, low weight and high cost-efficiency. But developments are going beyond mere weight reduction. Important changes have also taken place in the inside of transformers. A great deal has happened here. Modern cast-resin transformers no longer feature generally copper windings. Over the years, a pioneering technology has been developed using aluminum foil insulated with high-grade plastic film. Reminiscent of the way capacitors are manufactured, this technology has prevailed because the foil winding combines a simple technique with high electrical safety. The insulation is exposed to much lower electrical stress than in other types of winding.

Whereas turn-to-turn voltages in a conventional round-wire winding can add up to twice the interlayer voltage, the interlayer voltage of a foil winding never exceeds the single voltage per turn because all layers uniquely consist of a single winding which ensures excellent power- frequency voltage and impulse voltage strength.

But why aluminum? The coefficients of expansion of aluminum are very similar to those of the Siemens recipes for casting resin and result in only minimal mechanical stresses particularly in connection with load-change-induced temperature fluctuations and the associated dimensional changes. Moreover, efficiency calculations based on a holistic evaluation of conductivity, material price, specific weight and loss balance demonstrated that the aluminum conductor invariably offers advantages.

Considerable know-how and experience needed

Windings are fabricated using modern strip winding machines which enable several high-voltage and low-voltage phase windings to be produced in one shift. Once the primary and secondary-side windings have been completed, the coils are casted respectively sealed with a special resin mixture. The cast epoxy resin used to encapsulate the high-voltage coils is applied under vacuum and at high temperature. This process prevents harmful gas inclusions. Special vacuum pumps generate the necessary vacuum for this process in the casting kettle.

This process also requires expertise and experience if the desired objective is to be achieved. Principal Key Expert Hofmann knows that despite good pumps, controlling the

casting process is just as important for void-free casting. Under less than one millibar of pressure, a six cubic meter-capacity casting chamber still contains around six liters of its original 6000-liter air content. This residual air is enough to make life difficult for those in charge. That's why expertise in casting is crucial.

One of the secrets is that vacuum conditions in the casting chamber must be the same as or "poorer" than in the formulating area. Any residual air remains in the compound despite the high-vacuum conditions, and undesired bubbles are unable to form.

The quality of the encapsulation in conjunction with the electrical advantages of the foil winding is a crucial factor for ensuring freedom from partial discharges. Since this critical criterion applies throughout the life of cast-resin transformers, Siemens in Kirchheim tests every GEAFOL transformer at twice the rated voltage and stipulates much more stringent release criteria for the permitted partial-discharge inception level than those specified in European and international standards.

The core is comprised of grain-oriented, cold-rolled electric sheet steel whose thickness and layering are a function of requirements mainly related to no-load losses and noise. The Kirchheim engineers manufacture the transformer cores using several automatic core winding machines – the most modern of their kind.

Using vacuum switches with GEAFOL transformers

Contrary to standard distribution networks, the transformers installed in wind turbines will be switched off and on at MV side more frequently, which results in additional stress of the insulation. Switches must control the switching of transformers reliably and safely, with no need for overvoltage protection.

An important parameter in transformers is the magnetization current, one of the "small inductive currents". Interrupting these currents naturally creates marked transients but no unacceptably high switching overvoltages that would pose a threat to connected distribution transformers are permitted.

Extensive trials using a combination of Siemens GEAFOL transformers and vacuum switches have proven that the GEAFOL medium-voltage windings can handle switching overvoltages with no difficulty - providing further proof of their high product quality and operational safety.

Higher power ratings required

Like rolling mills or high-tech factories, larger wind turbine generators increasingly require dry-type transformers with higher ratings. Reasons for this are that cast-resin transformers ensure security of supply because they are highly reliable in operation and they are also extremely efficient, so helping to reduce operating costs (OPEX).

Higher power ratings in the case of cast-resin transformers generally also mean higher voltages and currents and consequently bigger heat loss. The development engineers in the Swabian town Kirchheim/Teck have devised a solution to the continuous problem of heat. For a long time now they have been using special cooling ducts for transformers rated above 630 kVA. However, these have previously only been used in conjunction with the low-voltage winding and have so far never featured in the fully encapsulated high-voltage winding.

To achieve very high MVA ratings, however, additional cooling ducts have to be incorporated both on the high-voltage side and inside the core.

As already mentioned, encapsulating the transformers generally calls for a great deal of expertise and experience. To ultimately permit production of a first-class, reliable cast-resin transformer with optimum electrical properties it was ultimately also necessary to modify the encapsulation process for transformers designed to incorporate a cooling duct in the high-voltage winding, and several hurdles had to be overcome with regard to optimizing the manufacturing process. The design of powerful cast-resin transformers is based on the experience of Siemens design and development specialists, accumulated over many decades and the use of high-tech design analysis tools and production equipment. Principal Key Expert Bernd Hofmann and his skilled Team have now got to grips with the design and the global casting process for GEAFOL transformers rated for up to about 50 MVA. The whole concept is running so smoothly now that the electrical properties of the resin insulation can be described as optimum. In the words of Bernd Hofmann: "The quality of the encapsulation in conjunction with the electrical advantages of the foil winding is crucial to ensure freedom from partial discharges – at least up to twice the rated voltage. Ultimately, it is the optimized combination of aluminum foil and strip windings and the specially coordinated cast-resin insulation that makes GEAFOL transformers reliable, safe, flame-retardant, self-extinguishing, practically maintenance-free and non-toxic."

The largest cast-resin transformers that have been built in Kirchheim to date are rated at 40 MVA (AN) and 50 MVA have already been designed. Also, a number of 23 MVA GEAFOL transformers are providing reliable service in a power plant in Hamburg.

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